

LECTURE 9

PANEL DATA MODELS (1)

(A) Introduction to Panel Data

(B) Unobservable Heterogeneity and an Example of Identical Twins

(C) First Differenced Equation and its Estimators

(A) Introduction to Panel Data

- A panel data set has both cross-sectional and time series dimensions. To collect panel data -- sometimes called longitudinal data -- we follow (or attempt to follow) the same group of individuals, families, firms, cities, states, or whatever, across time. For example, we collect the information of a same group of people's expenditure, income, education, wealth and study their consumption behavior.

Recall: (i) A cross sectional data: $i = 1, 2, \dots, 526$; $n = 526$

A Cross-Sectional Data Set on Wages and Other Individual Characteristics

<i>obsno</i>	<i>wage</i>	<i>educ</i>	<i>exper</i>	<i>female</i>	<i>married</i>
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

(ii) A time series data: $t = 1948, 1949, \dots, 1996$; $T = 49$ yearly observations

Partial Listing of Data on U.S. Inflation and Unemployment Rates, 1948–1996

Year	Inflation	Unemployment
1948	8.1	3.8
1949	−1.2	5.9
1950	1.3	5.3
1951	7.9	3.3
⋮	⋮	⋮
1994	2.6	6.1
1995	2.8	5.6
1996	3.0	5.4

(iii) Panel data: $i = 1, 2, \dots, 150$, and $t = 1986, 1990$; the total # of observations is $150 \times 2 = 300$

A Two-Year Panel Data Set on City Crime Statistics

<i>obsno</i>	<i>city</i>	<i>year</i>	<i>murders</i>	<i>population</i>	<i>unem</i>	<i>police</i>
1	1	1986	5	350000	8.7	440
2	1	1990	8	359200	7.2	471
3	2	1986	2	64300	5.4	75
4	2	1990	1	65100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
297	149	1986	10	260700	9.6	286
298	149	1990	6	245000	9.8	334
299	150	1986	25	543000	4.3	520
300	150	1990	32	546200	5.2	493

The first two rows are the two years of observations for city 1.

Example: Return to Education

Cross sectional data:

$$\ln(wage_i) = \beta_0 + \beta_1 educ_i + \beta_2 exper_i + u_i, \text{ where } i = 1, \dots, n$$

These n people (i.e., n observations) are randomly selected from a population at a given point of time.

Panel data:

$$\ln(wage_{it}) = \beta_0 + \beta_1 educ_{it} + \beta_2 exper_{it} + u_{it},$$

where $i = 1, \dots, n; t = 1, 2, \dots, T$.

These n people (i.e., n observations) are randomly selected from a population at a given point of time. Then, these same n people are re-interviewed at several subsequent points in time.

Example: Relationship between *crime rate* and *number of police officers*

Cross sectional data:

$$crime_rate_i = \beta_0 + \beta_1 number_of_police_i + u_i, \text{ where } i = 1, \dots, n$$

These n cities (i.e., n observations) are selected from a population at a give point of time, say year 2009.

Panel data:

$$crime_rate_{it} = \beta_0 + \beta_1 number_of_police_{it} + u_{it},$$

where $i = 1, \dots, n; t = 2009, 2010, 2011, 2012$

First we collect the data in 2009 and then we recollect the data of the crime rate and number of police officers from the same n cities in the next three following years.

(B) Unobservable Heterogeneity and an Example of Identical Twins

- Consider a multiple regression model as follows.

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i, \text{ where } i = 1, \dots, n$$

When x_1 endogenous, $Cov(x_1, u) \neq 0$, all OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ are biased and inconsistent. OLS is unreliable.

- One example of endogeneity is due to omitted variables in the error term. In the example of return to education, *ability* is an important unobservable individual specific variable absorbed by the error. We use OV to denote this unobservable variable, i.e., $u = \gamma OV + \varepsilon$. When $Cov(x_1, OV) \neq 0$, in general

$$Cov(x_1, u) = \gamma Cov(x_1, OV) + Cov(x_1, \varepsilon) \neq 0$$

- Here, OV is the unobserved heterogeneity because this individual specific (heterogeneity) variable may explain a lot of the variation of y , but unobserved.

- Some examples of unobserved heterogeneity are:
 - (i) In the empirical studies on labor supply, or student's performance using household survey, OV could be household specific factors, e.g., habit, family tradition, or even dinner menu, etc., which are not observed in the data.
 - (ii) In studies of economic growth using country-level data, OV could be endowment (e.g., oil), emphasis on education, work ethic, culture, etc.
- Unobserved heterogeneity is considered as one important source of endogeneity:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \gamma OV_i + \varepsilon_i$$

Solutions: we need to control or remove OV . Additional information is needed

(a) Proxy Variables

(b) Instrumental Variables (IV)

(c) panel data

Example: We could use (identical) twins to solve for the unobserved heterogeneity problem.

$$\ln(\text{wage}) = \beta_0 + \beta_1 \text{educ} + u = \beta_0 + \beta_1 \text{educ} + \gamma \cdot \text{ability} + \varepsilon$$

where $u = \gamma \cdot \text{ability} + \varepsilon$. Due to the fact that unobserved heterogeneity *ability* is correlated with *educ*, this *educ* is endogenous. OLS estimation is no longer reliable.

In the last two weeks, we discussed how to deal with the endogeneity problem by using IV estimators. (Recall: 2SLS estimation)

Another way of getting rid of unobservable ability is using identical twins proposed by Ashenfelter and Krueger (1994). They interviewed 495 separated twins over the age of 18.

Basic idea: Identical twins should have the same underlying ability. So, the ability can be differenced away by using *twin* differences.

- Suppose we have many identical twins (or pairs) $(T_{i1}, T_{i2}), i = 1, \dots, n$.

For each identical twin,

$$\text{First:} \quad \ln(\text{wage}_{i1}) = \beta_0 + \beta_1 \text{educ}_{i1} + \gamma \cdot \text{ability}_i + \varepsilon_{i1},$$

$$\text{Sibling:} \quad \ln(\text{wage}_{i2}) = \beta_0 + \beta_1 \text{educ}_{i2} + \gamma \cdot \text{ability}_i + \varepsilon_{i2},$$

where the *ability* is assumed to be identical.

- Differencing the first from the second:

$$\ln(\text{wage}_{i2}) - \ln(\text{wage}_{i1}) = \beta_1(\text{educ}_{i2} - \text{educ}_{i1}) + (\varepsilon_{i2} - \varepsilon_{i1})$$

The underlying *ability* is differenced away.

- The new explanatory $\Delta \text{educ}_i = \text{educ}_{i2} - \text{educ}_{i1}$ is exogenous. Therefore, the OLS regression above is consistent.

$$\begin{aligned} \ln(\text{wage}_{i2}) - \ln(\text{wage}_{i1}) &= \beta_1(\text{educ}_{i2} - \text{educ}_{i1}) + (\varepsilon_{i2} - \varepsilon_{i1}) \\ \Rightarrow \quad \Delta \ln(\text{wage}_i) &= \beta_1 \Delta \text{educ}_i + \Delta \varepsilon_i \end{aligned}$$

(C) First Differenced Equation and its Estimators

- The previous twins example of using two observations for each pair to wipe out the unobservable heterogeneity (*ability*) which could be correlated with the explanatory variable (education). Thus, it solves the endogeneity problem.

- In general, we can use **panel data** models: for $i = 1, \dots, n$

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it} = \beta_0 + \beta_1 x_{it} + a_i + v_{it},$$

- additional subscripts: $t = 1, 2$
- panel data: data set contains a time series for *each* cross-sectional unit
- the **composite error** term $u_{it} = a_i + v_{it}$, where a_i captures all *unobserved*, time-constant factors that affect y_{it} . a_i is called **fixed effect** (fixed over time) or unobserved heterogeneity, which could be correlated with x_{it}
- v_{it} : time-varying **idiosyncratic error**, representing unobserved factors that change over time and affect y_{it} . (Note: v_{it} is same as ε_{it})
- advantage of panel data: control for fixed effects/unobserved heterogeneity

Example: City crime in US cities in year 1986 and again 1990

The first two rows are the two years of observations for city 1.

A Two-Year Panel Data Set on City Crime Statistics

<i>obsno</i>	<i>city</i>	<i>year</i>	<i>murders</i>	<i>population</i>	<i>unem</i>	<i>police</i>
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- Famous examples of panel data sets:

- (i) Panel Study of Income Dynamics (PSID) in the US: households

- over 18,000 individuals living in 5,000 families in the US since 1968
 - The data have been collected continuously, including employment, income, wealth, expenditures, health, marriage, childbearing, etc. It is the longest running household panel survey

- (ii) Penn World Table: countries studies

- international prices for 167 countries for some or all of the years 1950-2014
 - GDP, inflation, population, and other variables
 - research on development and growth

- Since a_i is unobserved and possibly correlated with x_{it} , OLS of regressing y on x could be inconsistent due to omitted variable bias.

- **First-differenced equation**

The panel data model: $y_{it} = \beta_0 + \beta_1 x_{it} + a_i + v_{it}$

$$t = 1: \quad y_{i1} = \beta_0 + \beta_1 x_{i1} + a_i + v_{i1}$$

$$t = 2: \quad y_{i2} = \beta_0 + \beta_1 x_{i2} + a_i + v_{i2}$$

We subtract the first equation from the second equation and obtain

$$y_{i2} - y_{i1} = \beta_1 (x_{i2} - x_{i1}) + (v_{i2} - v_{i1})$$

or $\Delta y_i = \beta_1 \Delta x_i + \Delta v_i$, called the **first-differenced equation**, a single cross-sectional equation with variables differenced over time.

- After the first difference, we remove the unobservable heterogeneity and thus solve the endogeneity problem.

- If v_{it} is uncorrelated with x_{i1} and x_{i2} , then Δv_i is uncorrelated with Δx_i . Under this assumption, the OLS estimator of β_1 from the first-differenced equation is called the **first-differenced (FD)** estimator. If all the classical linear model assumptions are satisfied, then FD estimators are unbiased, consistent and all the statistical inference is exact as we learned in the last semester.
- The cost of using FD estimation:
 - ✚ Cost 1: Panel data sets are harder to collect than a single cross section or pooled cross sections, especially for individuals.
 - ✚ Cost 2: The effect of time-invariant explanatory variables cannot be identified since they are absorbed into the heterogeneity effect a_i .
 - Panel data model with time-invariant observable variables:

$$y_{it} = \beta_0 + \beta_1 x_{it} + \beta_2 z_i + a_i + v_{it}, t = 1, 2$$

z_i : time-invariant *observable* variables. The gender, ethnic, location, etc. are also removed with a_i when we get the first-differenced equation.

That is

$$t = 2: y_{i2} = \beta_0 + \beta_1 x_{i2} + \beta_2 z_i + a_i + v_{i2}$$

$$t = 1: y_{i1} = \beta_0 + \beta_1 x_{i1} + \beta_2 z_i + a_i + v_{i1}$$

Once we subtract the first equation from the second, we obtain

$$y_{i2} - y_{i1} = \beta_1(x_{i2} - x_{i1}) + (v_{i2} - v_{i1})$$
$$\Rightarrow \beta_2 \text{ cannot be estimated. } \underline{\text{It's gone}}.$$

Example: Suppose we do not have the twins dataset. But, we interview the same group of random selected individuals in year 1 and year 2.

Consider $\ln(wage_{it}) = \beta_0 + \beta_1 educ_{it} + \beta_2 female_i + a_i + v_{it}, t = 1, 2$

$$\Delta \ln(wage_{it}) = \beta_1 \Delta educ_t + \Delta v_i$$

We cannot estimate partial effect of female on wage, i.e., β_2 cannot be estimated.

We might not be able to estimate partial effect of *educ* on wage as well. Why?

Example: The effect of alcohol taxes on traffic deaths in the U.S.

(Panel_beertax.dta)

Some facts

- Approximately 40,000 traffic fatalities annually in the U.S.
- 33% of traffic fatalities involve a drinking driver
- 25% of drivers on the road between 1am and 3am have been drinking
- A drunk driver is 13 times as likely to cause a fatal crash as a non-drinking driver (estimate)

A (balanced) panel data set: 48 U.S. states over 7 years (1982-1988) with variables:

(“balanced” means each state has the same number of observations over time)

- *vfrall*: traffic (vehicle) fatality rate (# of traffic deaths per 10,000 state residents)
- *BeerTax* : tax on a case of beer
- Other variables (legal driving age, drunk driving laws, etc.)

$$vfrall_i = \beta_0 + \beta_1 BeerTax_i + u_i$$

We examine how taxing beer could reduce traffic deaths. Parameter of interest: β_1

(1) We use the observations in year1988 to do regression analysis. For this case, it is just a usual cross-sectional regression model.

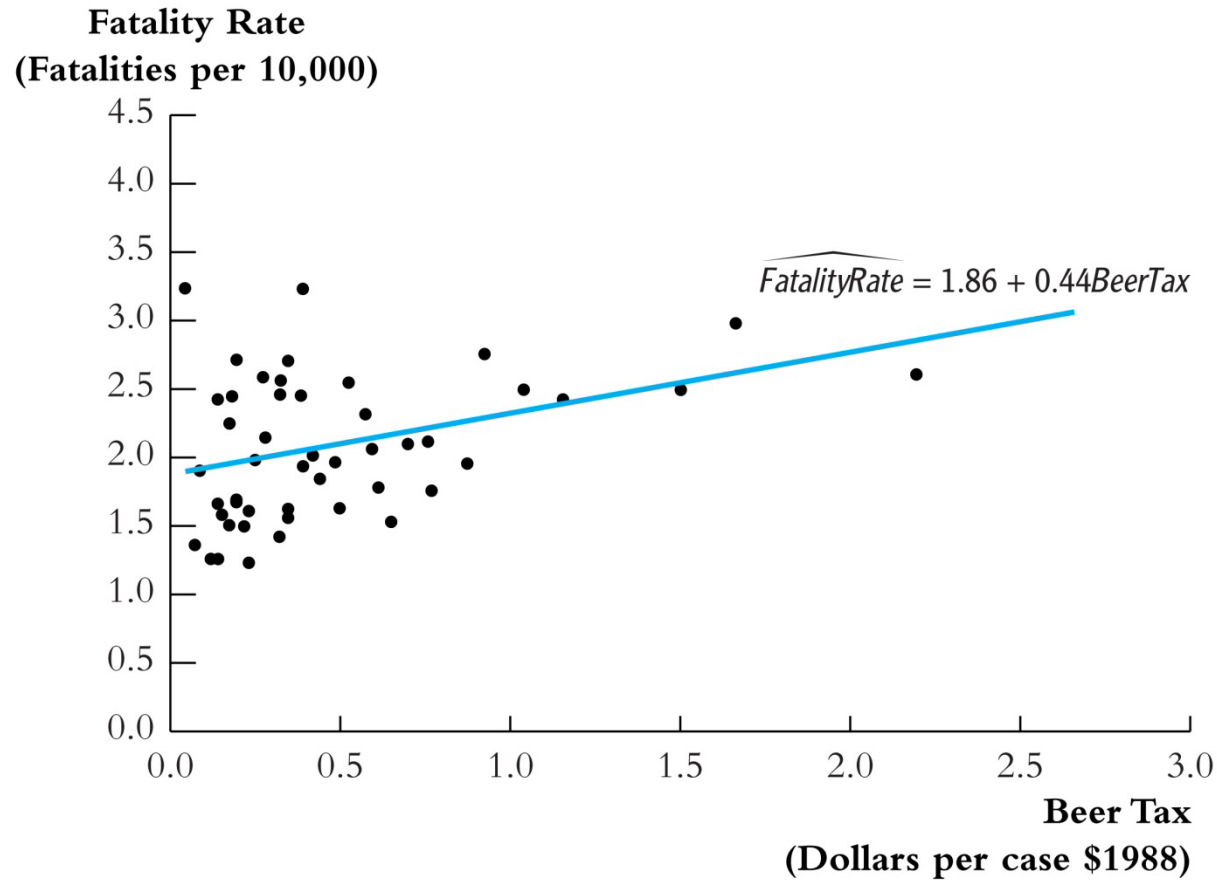
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. reg vfrall beertax if (year==1988)
```

Source	SS	df	MS	Number of obs = 48		
Model	1.71077215	1	1.71077215	F(1, 46) = 7.12		
Residual	11.0559133	46	.24034594	Prob > F = 0.0105		
				R-squared = 0.1340		
				Adj R-squared = 0.1152		
Total	12.7666854	47	.271631604	Root MSE = .49025		

vfrall	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
beertax	.4387546	.1644538	2.67	0.011	.1077262	.769783
_cons	1.859073	.1059887	17.54	0.000	1.645729	2.072417

The estimated slope is positive. It means higher alcohol taxes lead to more traffic deaths. Does it make sense?

If yes, the policy implication should be to cut the beer tax (and encourage people to drink more beers.)



There is a positive relationship between *Deaths* and *Beer Tax*. Clearly, this cannot be the case in the reality. The OLS estimate is clearly wrong and misleading.

(2) “Pooled” OLS (regressing *vfrall* on *beertax* using all observations):

```
. reg vfrall beertax
```

Source	SS	df	MS	Number of obs = 336		
Model	10.1686586	1	10.1686586	F(1, 334) = 34.39		
Residual	98.7468513	334	.295649255	Prob > F = 0.0000		
Total	108.91551	335	.325120925	R-squared = 0.0934		
				Adj R-squared = 0.0906		
				Root MSE = .54374		

vfrall	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
beertax	.3646054	.0621698	5.86	0.000	.2423117	.4868992
_cons	1.853308	.0435671	42.54	0.000	1.767607	1.939008

This is still counterintuitive. Something must be very wrong. One explanation could be that people impose higher beer tax in states with more traffic deaths. The positive correlation between *vfrall* and *beertax* reflects this reverse causality.

Alternatively, important factors that determine traffic deaths may be omitted. The state specific factors (α_i) could be the important omitted variable in the previous regression analysis.

(3) Now we try to use the panel data to control for state effects.

$$vfrall_{it} = \beta_0 + \beta_1 beertax_{it} + a_i + v_{it}$$

a_i : state effects, modeling road condition, weather, driving culture, etc.

We use two time periods (1982 & 1988) from the dataset. The first differenced equation is as follows.

$$\Delta vfrall_i = \beta_1 \Delta beertax_i + \Delta v_i$$

The FD estimates are (Panel_beertax8288.dta)

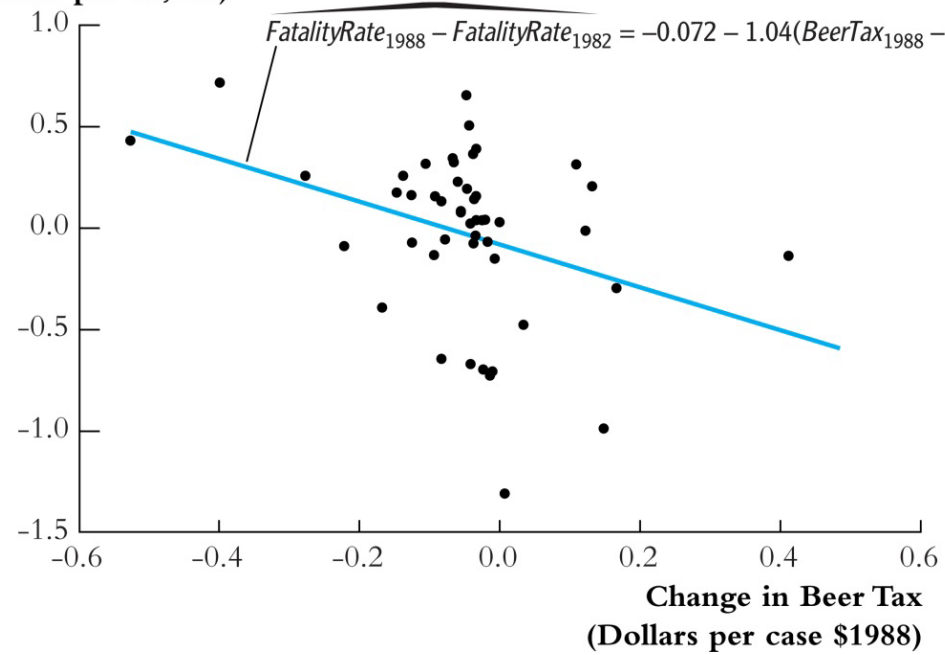
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. reg dvfrall dbeertax
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Source	SS	df	MS	Number of obs = 48		
Model	.966447698	1	.966447698	F(1, 46) =	6.22	
Residual	7.14175382	46	.155255518	Prob > F =	0.0162	
Total	8.10820152	47	.172514926	R-squared =	0.1192	
				Adj R-squared =	0.1000	
				Root MSE =	.39402	

dvfrall	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
dbeertax	-1.040972	.417228	-2.49	0.016	-1.880809	-.2011357
_cons	-.072037	.060644	-1.19	0.241	-.1941071	.0500331

Taxing on beer reduces the traffic deaths after controlling for (i.e., removing) state effects!

**Change in Fatality Rate
(Fatalities per 10,000)**



Example: We use the Crime2.dta data set that consists of data from 46 cities in years 1982 and 1987.

Cross sectional model:

$$\text{Crime_rate}_i = \beta_0 + \beta_1 \text{unemployment_rate}_i + (a_i + v_i), \text{ where } i = 1, 2, \dots, n$$

Stata> reg crmrte unem if d87==1

(Note that “if d87==1” means we only use the year 1987 data)

Source	SS	df	MS			
Model	1775.90927	1	1775.90927	Number of obs =	46	
Residual	52674.6465	44	1197.15106	F(1, 44) =	1.48	
Total	54450.5558	45	1210.01235	Prob > F =	0.2297	
				R-squared =	0.0326	
				Adj R-squared =	0.0106	
				Root MSE =	34.6	

crmrte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
unem	-4.161134	3.416456	-1.22	0.230	-11.04655	2.72428
_cons	128.3781	20.75663	6.18	0.000	86.54589	170.2104

The $\hat{\beta}_1$ is negative 4.161, though it is not significant. It is counterintuitive; we expect to have a negative sign for β_1 .

First-differenced equation:

$$\Delta \text{Crime_rate}_i = \beta_1 \Delta \text{unemployment_rate}_i + \Delta v_i$$

```
. reg ccrmrte cunem
```

Source	SS	df	MS	Number of obs = 46		
Model	2566.43744	1	2566.43744	F(1, 44) = 6.38		
Residual	17689.5497	44	402.035219	Prob > F = 0.0152		
Total	20255.9871	45	450.133047	R-squared = 0.1267		
				Adj R-squared = 0.1069		
				Root MSE = 20.051		

ccrmrte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
cunem	2.217999	.8778658	2.53	0.015	.4487771	3.987222
_cons	15.4022	4.702117	3.28	0.002	5.92571	24.8787

Note that one of the costs of using panel here is we are not able to estimate the partial effect of fixed observable variables, for example gender, location, culture, etc., on the change of the crime rate.